

Benchmarking Local Large Language Models for Structured Knowledge Extraction in Conversational Commerce

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Abstract

The integration of natural language understanding into conversational commerce requires robustly extracting structured intents from idiomatic text. While cloud-based models are highly capable, their latency, privacy risks, and operational costs necessitate efficient local alternatives. To address this, we empirically evaluate locally deployed, open-weights large language models (LLMs) on parsing multi-intent shopping cart commands into deterministic JSON schemas. We benchmark multiple models across minimal, extended, and few-shot prompting paradigms, utilizing a dual-axis framework to simultaneously measure extraction quality (operation F1-score, exact match, schema validity) and hardware performance (latency, GPU power consumption, VRAM usage). Our evaluation highlights a fundamental trade-off between schema compliance and computational overhead, demonstrating how advanced prompting techniques improve parsing accuracy while measurably impacting inference latency and power draw. Ultimately, this comprehensive profiling identifies optimal model configurations for resource-constrained environments, establishing a practical baseline for deploying cost-effective and energy-efficient LLMs in edge-based retail automation.

Keywords: LLM; Prompt Engineering; Natural Language Understanding; JSON Parsing; Structured Output; Model Performance; Conversational Agents; Information Extraction;

1. Introduction

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2. Background and Related Work

3. Methodology

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4. Results and Discussion

We evaluate six open-weights models under three prompting regimes—*minimal*, *extended*, and *few-shot*—yielding 18 model-prompt configurations, each applied to the full set of 5000 held-out shopping-cart commands. These commands contain a mean of 3.64 atomic operations (18,175 gold operations in total), so a single utterance typically bundles several add/remove intents that must be jointly recovered. Following the dual-axis protocol introduced in the Methodology, we report extraction quality (Table 1) and computational cost (Table 2) for every configuration, and then analyse the quality–cost trade-off that the prompting regime induces (Figure 5). Throughout, we use the operation-level micro-averaged F1 score (Op-F1) as the primary quality scalar and an estimated energy per command, $E \approx \bar{P} \bar{t}$ (mean GPU power $\times$ mean latency), as the primary cost scalar.

4.1. Extraction Quality and the Role of the Prompting Regime

Table 1. Extraction quality by model and prompting regime (Min, minimal; Ext, extended; FS, few-shot). JSON and Schema denote the JSON- and schema-validity rates; Op-F1 is the operation-level micro-averaged F1 score; EM is the ordered full-command exact-match rate; Act, Prod, and Qty are the action-, product-, and quantity-field accuracies. All values are computed over the 5000-command evaluation set. The best Op-F1 per model is shown in bold.

Model	Prompt	JSON	Schema	Op-F1	EM	Act	Prod	Qty
Gemma4 31B	Min	1.000	1.000	0.951	0.833	0.969	0.995	0.984
	Ext	1.000	1.000	0.967	0.885	0.978	1.000	0.989
	FS	1.000	1.000	0.981	0.933	0.981	1.000	1.000
GPT-OSS 20B	Min	1.000	1.000	0.923	0.753	0.968	0.991	0.958
	Ext	0.999	0.999	0.944	0.819	0.981	0.994	0.967
	FS	1.000	1.000	0.975	0.915	0.995	0.999	0.980
Llama3.1 8B	Min	0.935	0.881	0.782	0.538	0.856	0.808	0.853
	Ext	0.998	0.976	0.801	0.568	0.929	0.860	0.949
	FS	0.997	0.995	0.950	0.841	0.977	0.966	0.988
Mistral 7B	Min	0.963	0.877	0.689	0.357	0.842	0.762	0.855
	Ext	0.982	0.865	0.504	0.281	0.871	0.559	0.887
	FS	0.993	0.976	0.942	0.804	0.950	0.963	0.973
Granite4 3B	Min	1.000	0.928	0.797	0.512	0.930	0.871	0.930
	Ext	1.000	0.931	0.790	0.516	0.921	0.859	0.946
	FS	1.000	0.997	0.937	0.807	0.984	0.958	0.987
Llama3.2 3B	Min	0.967	0.606	0.610	0.311	0.758	0.700	0.749
	Ext	0.970	0.637	0.526	0.229	0.681	0.581	0.754
	FS	0.890	0.822	0.813	0.566	0.796	0.810	0.826

Few-shot prompting yields the best Op-F1 for all six models, and the magnitude of the gain is inversely related to the model’s zero-shot capability. Relative to the minimal prompt, in-context demonstrations raise Op-F1 by only +0.031 for Gemma4 31B and +0.051 for GPT-OSS 20B, but by +0.140 (Granite4 3B), +0.168 (Llama3.1 8B), +0.203 (Llama3.2 3B), and +0.253 (Mistral 7B) for the smaller models. The exemplars thus act as a capability equaliser: models that are mediocre zero-shot parsers become competitive once shown the target format, whereas the largest models—already near ceiling—have little headroom to gain. Figure 1 visualises this compression of the quality gap between small and large models.

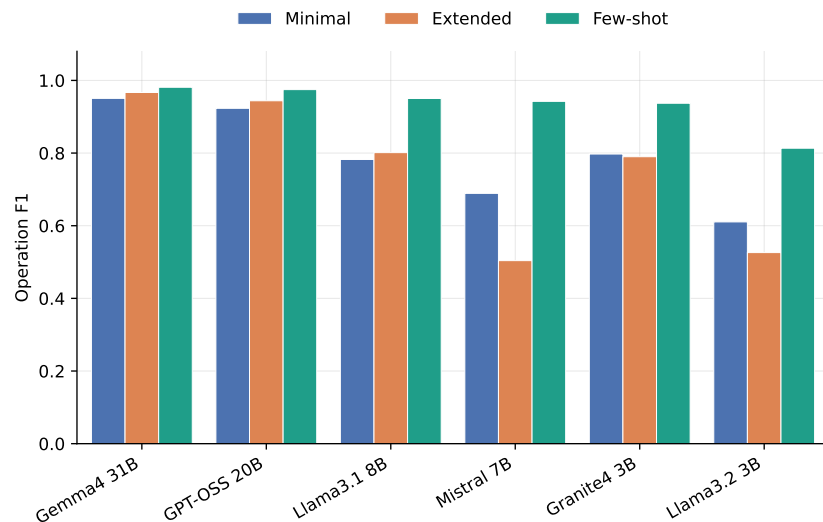


Figure 1. Operation-level F1 by model and prompting regime. Few-shot prompting (teal) yields the best accuracy for all six models and sharply compresses the gap between small and large models; the extended prompt (orange) is inconsistent, degrading Mistral 7B and the Llama3.2 3B small model relative to the minimal prompt (blue).

Decomposing quality into format adherence and content correctness clarifies the mechanism. Under the minimal prompt, several small models emit syntactically valid JSON that nonetheless violates the target schema: schema validity is only 0.928 (Granite4 3B), 0.881 (Llama3.1 8B), and 0.606 (Llama3.2 3B) despite JSON validity at or above 0.94. Few-shot demonstrations repair this format gap almost entirely—schema validity rises to 0.997, 0.995, and 0.822, respectively—indicating that the dominant failure mode of small models is structural (key naming, value typing, nesting) rather than semantic. Demonstrations specify the output contract by example rather than by description, which the smaller models follow more reliably than prose instructions. Figure 2 reports all seven quality metrics across models and regimes, making this format-versus-content decomposition visually explicit.

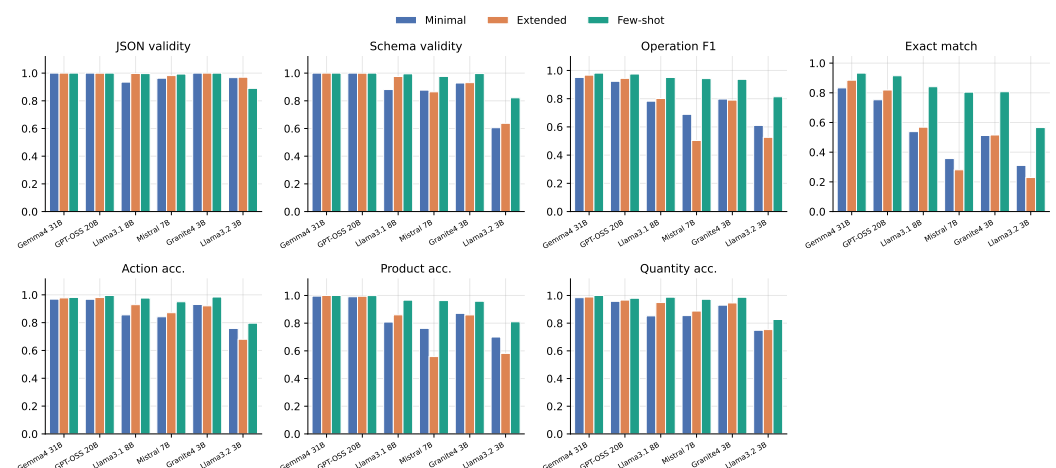


Figure 2. All quality metrics by model and prompting regime; each panel is a grouped bar chart over the six models (bars: minimal, extended, few-shot). Few-shot prompting (teal) is best across essentially every metric for the capable models, and the largest small-model gains appear on schema validity and exact match, confirming that the principal failure mode under weaker prompts is structural rather than semantic.

In contrast, the extended (instruction-rich) prompt is not a monotonic improvement and is frequently harmful. It helps only the most capable models—Gemma4 31B (+0.016),

GPT-OSS 20B (+0.021), and Llama3.1 8B (+0.019)—while degrading Mistral 7B (−0.185) and Llama3.2 3B (−0.084), and leaving Granite4 3B essentially unchanged (−0.007). Manual inspection traces the Mistral regression to over-normalisation: the additional instructions induce the model to lemmatise product spans (“toothpastes” → “toothpaste”, “quinoa bags” → “quinoa bag”) and to misresolve idiomatic quantities (“a pair” → 1), collapsing product-field accuracy from 0.762 to 0.559 even though the emitted JSON remains well-formed. Longer instructions therefore introduce degrees of freedom that smaller models resolve in unintended, ostensibly “helpful” ways that diverge from the verbatim-span ground truth. This is an important negative result: scaling prompt complexity without grounding it in exemplars can reduce extraction fidelity.

4.2. Computational Cost and the Decode-Bound Regime

Table 2. Computational cost by model and prompting regime. Lat_{med} , Lat_{mean} , and Lat_{p95} are latency percentiles (s); Thr is throughput (commands/s); Pwr is mean GPU power (W); E is the estimated energy per command (J, $E \approx \bar{P} \bar{t}$); P-tok and E-tok are the mean prompt- and output-token counts. Lower is better for all columns except throughput.

Model	Prompt	Lat_{med}	Lat_{mean}	Lat_{p95}	Thr	Pwr	E (J)	P-tok	E-tok
Gemma4 31B	Min	13.03	16.33	37.24	0.06	156	2555	146	600
	Ext	13.25	16.27	36.62	0.06	156	2545	197	599
	FS	10.10	11.71	19.87	0.08	156	1820	436	426
GPT-OSS 20B	Min	2.46	2.54	3.90	0.36	124	315	196	345
	Ext	2.44	2.52	3.95	0.36	124	313	248	341
	FS	2.39	2.51	3.89	0.37	126	315	455	339
Llama3.1 8B	Min	0.72	0.71	0.98	1.06	141	101	138	79
	Ext	0.64	0.66	0.93	1.12	141	94	191	72
	FS	0.58	0.56	0.71	1.26	140	79	402	57
Mistral 7B	Min	0.61	0.59	0.84	1.24	146	87	148	88
	Ext	0.58	0.56	0.78	1.30	147	82	200	82
	FS	0.50	0.49	0.64	1.42	143	70	467	70
Granite4 3B	Min	0.65	0.61	0.82	1.21	119	73	136	98
	Ext	0.53	0.52	0.81	1.35	118	62	189	81
	FS	0.41	0.39	0.51	1.64	118	46	400	56
Llama3.2 3B	Min	0.51	0.53	0.76	1.34	123	65	148	90
	Ext	0.50	0.49	0.63	1.42	124	61	201	80
	FS	0.40	0.39	0.48	1.67	122	47	412	56

The cost profile is governed by decoding rather than prefill. Although few-shot prompts are three- to fourfold longer on the input side (prompt tokens rise from ~ 140 to ~ 400 – 470), they consistently reduce wall-clock latency because the demonstrations suppress verbose generation: mean output length falls (e.g., $98 \rightarrow 56$ eval tokens for Granite4 3B and $600 \rightarrow 426$ for Gemma4 31B), and decode dominates latency in autoregressive inference. Consequently, few-shot mean latency is 17.7–35.6% lower than minimal for five of the six models (Granite4 3B −35.6%, Gemma4 31B −28.3%, Llama3.2 3B −27.2%, Llama3.1 8B −21.2%, Mistral 7B −17.7%), with GPT-OSS 20B effectively flat (−1.4%). The conventional intuition that richer prompts cost more compute is thus inverted here: by constraining the output, exemplars make inference simultaneously more accurate and faster. Figure 3 contrasts the prompt and output token counts that drive this effect.

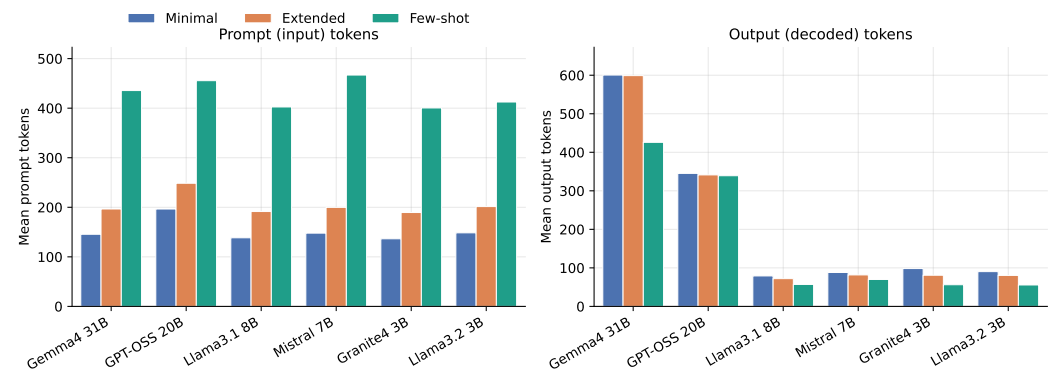


Figure 3. Mean prompt (input) and output (decoded) token counts per command, by model and prompting regime. Few-shot prompts are markedly longer on the input side (left) yet *shorter* on the output side (right): the demonstrations elicit concise, minified JSON and suppress verbose generation. Because decoding dominates latency, this reduction in output tokens is the mechanism behind the lower few-shot latency and energy reported in Table 2.

GPU power draw is essentially model-bound rather than prompt-bound—within each model it varies by only a few watts across regimes (e.g., ~156 W for Gemma4 31B, ~124 W for GPT-OSS 20B, ~118 W for Granite4 3B)—so the estimated energy per command tracks latency. Few-shot prompting therefore reduces per-command energy by 19–36% for the capable models, with the smallest models reaching 46 J (Granite4 3B) and 47 J (Llama3.2 3B) per command versus 315 J (GPT-OSS 20B) and 1820 J (Gemma4 31B). Energy spans a factor of roughly 40 across the frontier—a far wider dynamic range than quality—which is the crux of the deployment trade-off analysed in Section 4.3. Figure 4 summarises all eight cost metrics across the 18 configurations.

We report mean VRAM occupancy in the underlying logs for completeness but caution against ranking models by it: the measured footprint does not track parameter count (the 20B GPT-OSS model occupies only ~16 GB, less than the 3B Llama3.2 at ~17 GB or the 8B Llama3.1 at ~22 GB), which indicates that the figure reflects server-level allocator reservation and key–value-cache pre-allocation under the serving runtime rather than the minimal model footprint. Absolute VRAM values should accordingly be read as an upper bound on residency for this deployment, not as an intrinsic model property.

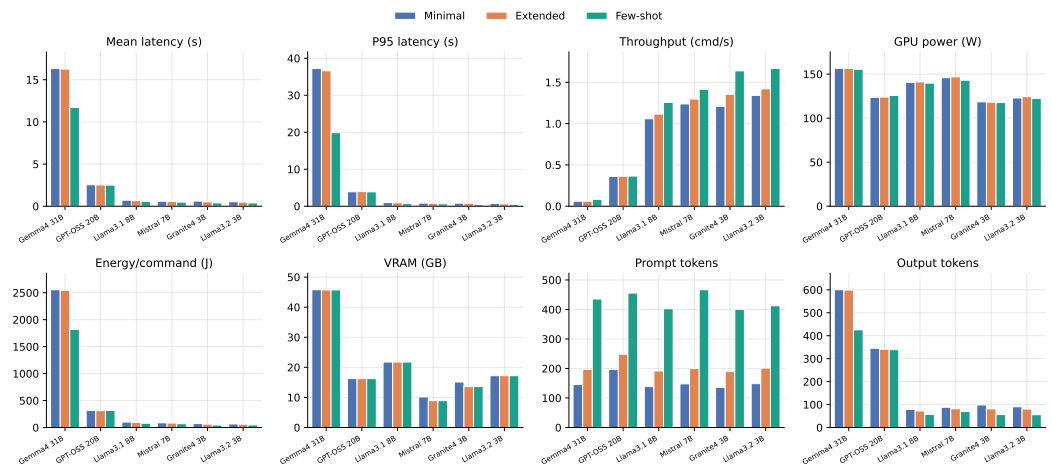


Figure 4. All cost metrics by model and prompting regime: mean and p95 latency, throughput, mean GPU power, estimated energy per command, VRAM occupancy, and mean prompt and output token counts. Latency, energy, and throughput improve under few-shot prompting for every model, whereas GPU power is essentially model-bound. The latency and energy panels are dominated by Gemma4 31B, reflecting the $\sim 40\times$ cost dynamic range across the frontier. VRAM reflects serving-runtime reservation rather than the minimal model footprint (see text).

4.3. The Quality–Cost Trade-off

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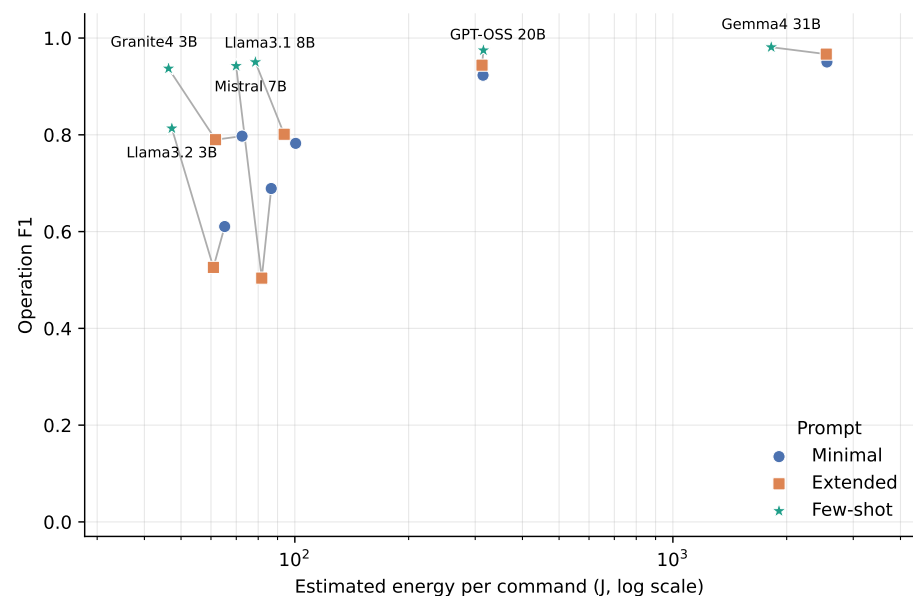


Figure 5. Quality–cost trade-off across all 18 configurations. Each point is one model–prompt configuration; thin grey lines connect the three regimes of a model. The horizontal axis is the estimated energy per command on a logarithmic scale. For every model the few-shot configuration (star) lies above and to the left of its minimal (circle) and extended (square) counterparts—simultaneously more accurate and more energy-efficient—so the operative trade-off is across model scale along the few-shot frontier rather than across prompts.

Figure 5 plots every configuration in the energy–quality plane and reveals the central finding of this study: for capable models, few-shot prompting is Pareto-improving rather than Pareto-trading—it moves each model up (higher F1) and to the left (lower energy) at once. The few-shot points dominate their own minimal and extended counterparts for all six models. The genuine trade-off is therefore not between prompts but across model scale: along the few-shot frontier, quality and cost are positively coupled, and the practitioner’s

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decision reduces to selecting an operating point on that frontier. Three operating points are practically distinct:

- **Maximum accuracy.** Gemma4 31B (few-shot) attains the highest fidelity (Op-F1 0.981, exact match 0.933) but at 1820 J and 11.7 s per command—appropriate only where accuracy is paramount and latency and energy are unconstrained, such as offline or server-side processing.
- **Balanced.** GPT-OSS 20B (few-shot) reaches Op-F1 0.975 at 315 J and 2.5 s, a $5.8\times$ energy reduction for a 0.6-point F1 loss relative to Gemma4 31B.
- **Efficiency frontier.** Granite4 3B (few-shot) achieves Op-F1 0.937—95.5% of Gemma4 31B's quality—at 46 J and 0.39 s, that is, roughly $40\times$ less energy and $30\times$ lower latency. Llama3.1 8B (0.950, 79 J) and Mistral 7B (0.942, 70 J) offer marginally higher accuracy at modestly higher cost.

Quality saturates rapidly with scale once few-shot prompting is applied: moving from a 3B to a 31B model improves Op-F1 by only ~ 0.044 while increasing energy by a factor of roughly 40. For the resource-constrained, edge-oriented retail setting that motivates this work, a 3–8B model with few-shot prompting is therefore the rational default, reserving 20–31B models for accuracy-critical or server-side deployments. Crucially, this recommendation is contingent on the prompt: the same small models are not viable under minimal or extended prompting (Op-F1 0.50–0.80), so the deployment win arises from the prompt–model pairing rather than from the model alone. The same Pareto structure holds when mean latency replaces energy on the cost axis (Figure 6), confirming that the ranking of configurations is robust to the choice of cost metric.

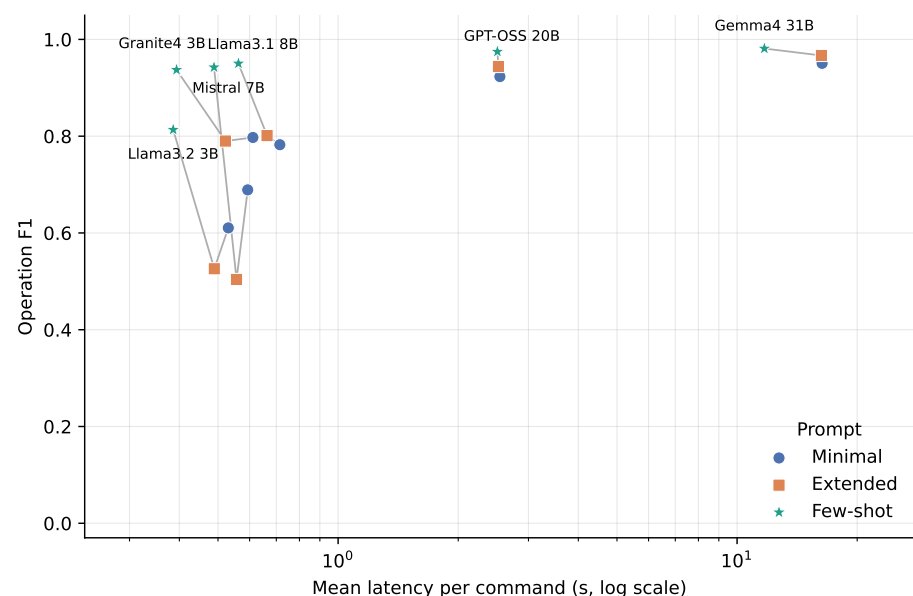


Figure 6. Quality–cost trade-off with mean latency per command (log scale) as the cost axis, complementing Figure 5. Markers denote the prompting regime and thin lines connect the three regimes of each model. The few-shot frontier again lies up and to the left and the model ordering is preserved, confirming that the conclusions are robust to whether cost is measured as latency or energy.

4.4. Error Analysis

Field-level accuracies localise the residual errors. Product-span extraction is the hardest field for every model that makes appreciable errors: under the minimal prompt the 3–8B models score only 0.70–0.87 on the product field, below both their action (0.76–0.93) and quantity (0.75–0.93) accuracies, reflecting sensitivity to surface form such as pluralisation; quantity resolution is additionally stressed by idiomatic numerals (“a pair”,

“half a dozen”, “a dozen”) that must be mapped to integers. For the near-ceiling large models the three fields all exceed 0.95 and their differences are negligible. The gap between operation-level F1 and whole-command exact match is informative: even Gemma4 31B (few-shot) scores 0.981 Op-F1 but 0.933 exact match, because a command counts as correct only if all of its 3.64 operations on average are simultaneously correct, so per-operation errors compound multiplicatively at the command level. Exact match is consequently the more stringent and deployment-relevant metric for transactional settings, where a single misparsed item corrupts the entire cart update.

5. Limitations and Future Work

6. Conclusions

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

LLMs	Large Language Models
NLP	Natural Language Processing
PEFT	Parameter-Efficient Fine-Tuning
LoRA	Low-Rank Adaptation
IA ³	Infused Adapter by Inhibiting and Amplifying Inner Activations
ICL	In-Context Learning
JSON	JavaScript Object Notation
SFT	Supervised Fine-Tuning
F1	F1 Score
GPU	Graphics Processing Unit
DDP	Distributed Data Parallel
MLP	Multi-Layer Perceptron
bf16	BFloat16 Precision Format
tf32	TensorFloat-32

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